



THE 12th INTERNATIONAL SYMPOSIUM ON EAST ASIAN RESOURCES RECYCLING TECHNOLOGY



3rd-7th November, 2013
Zhangjiajie · Hunan · China

Organizers



The Nonferrous Metal Society of China



Central South University

■ Co-Organizers

- Chinese National Engineering Research Center for Control & Treatment of Heavy Metal Pollution
- The Mining and Materials Processing Institute of Japan
- The Korean Institute of Resources Recycling
- The Formosa Association of Resource Recycling
- Energy Research Institute of Chulalongkorn University, Thailand

■ Support

- Ministry of Science and Technology of the People's Republic of China
- Ministry of Environmental Protection of the People's Republic of China
- National Natural Science Foundation of China
- China Nonferrous Metals Industry Association
- China Association of Resource Comprehensive Utilization
- Professional Committee for Pollution Prevention & Control of Heavy Metal of Chinese Society for Environmental Sciences
- International Copper Association Ltd., China
- New World Environmental Services Group
- Hunan Vary Tech Co., Ltd

Welcome to the 12th International Symposium on East Asian Resources Recycling Technology (EARTH2013)

On behalf of the organizers of the 12th International Symposium on East Asia Resources Recycling Technology (EARTH2013), the Nonferrous Metals society of China (NFsoc) and Central South University, I sincerely invite you to attend EARTH2013 to be held in Zhangjiajie, China on November 3-7, 2013.

After over 20 years of development, the International Symposium on East Asia Resources Recycling Technology (EARTH) has become one of the oldest conferences that have the largest coverage and the most influence in the field of recycling technology in East Asia. Following the successful eleven predecessors of the EARTH series, EARTH2013 will cover seven major topics and share the experiences and achievements related to all aspects of resources recycling in the last two years. As usual we will invite representatives of member countries and regions to deliver plenary lectures in the opening ceremony, which will mainly focus on the latest and most important research achievements in resource recycling technology in these countries and regions. EARTH2013 will bring some new features. The topics will emphasize on seeing resource recycling from the perspective of environmental protection and concentrate on two major issues, “urban mine” and “treatment and reclamation of heavy metals pollutants”. In addition, keynote speeches by distinguished young scientist will be a newly added section that will provide them with opportunities to present their latest research findings. EARTH2013 will be not only a unique opportunity for participants to learn the up-to-date advances at the forefront of this field, but a fabulous platform for industry experts, scholars, students, business people and officials to communicate and cooperate as well.

EARTH2013 will be held in Zhangjiajie, a beautiful eco tourist city. Located in the northwestern part of Hunan province and lying at the heart of Wuling Mountains, Zhangjiajie is one of the prime tourist destinations in China. With its long history and impressive views, Wulingyuan, a UNESCO World Heritage Site as well as a national AAAAA scenic area, is comprised of Zhangjiajie National Forest Park, Suoxi Valley Nature Reserve and Tianzishan mountain ranges. Zhangjiajie is largely inhabited by ethnic minorities, making it a splendid combination of rich ethnic customs and exquisite natural scenery. It has been striving to develop green tourism and aiming at the harmony between man and nature and sustainable development, perfectly matching the theme of the symposium of this year.

Participants all over the world are warmly welcomed to EARTH2013 in Zhangjiajie to celebrate the event together!

Liyuan Chai 

Ph.D, Professor

Chairman of EARTH2013

Topics

The scope of the symposium covers all aspects related to the resources recycling. Topics of interest include, but are not limited to the followings:

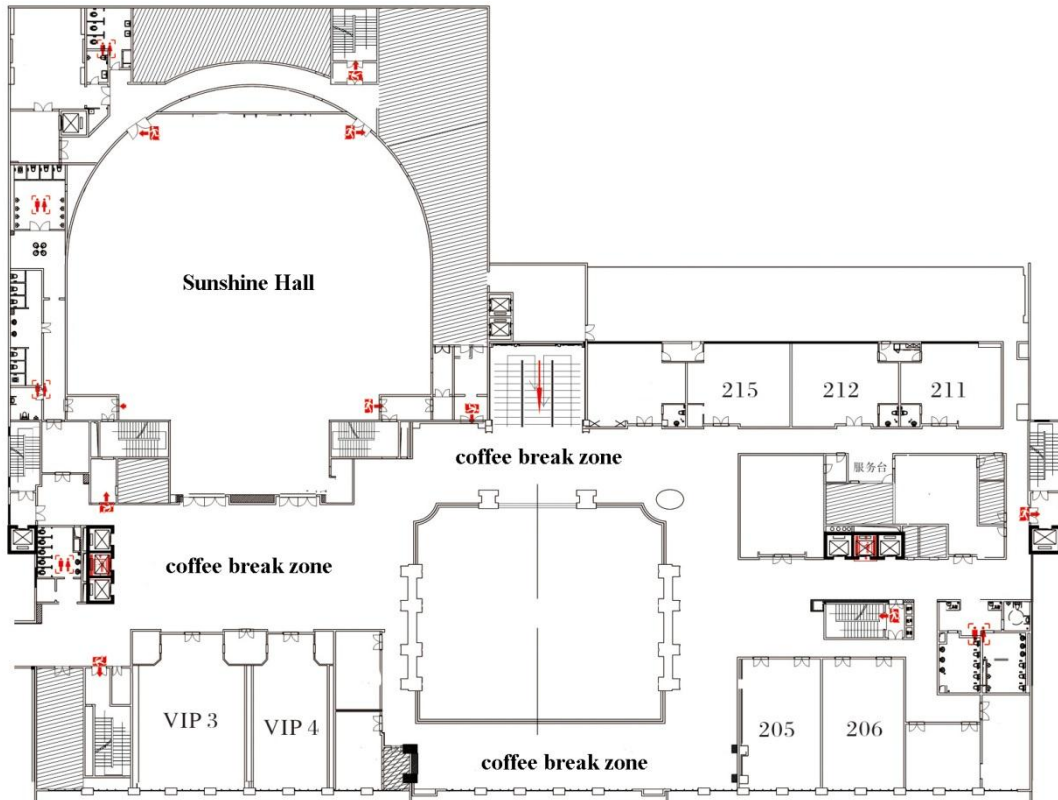
1. Policy, management system and life-cycle evaluation;
2. Recycling & utilization of industrial wastes: metals, slag, sludge, ash, catalyst, effluence, wastewater, gaseous waste, etc;
3. Recycling & utilization of electrical and electronic wastes;
4. Recycling & utilization of end-life vehicles, rubber, plastic, paper, glass and other living wastes;
5. Development of technologies for recycling and waste minimization production;
6. Treatment of heavy metals pollutants and resource recycle technology;
7. Environmental protection and other related issues.

Agenda

Time	Content	Venue
3 rd Nov., 2013		
13:00-18:00	Registration	Lobby of the first floor
18:00-21:00	Welcome party	Sunshine Hall
4 th Nov., 2013		
7:00-8:00	Breakfast	Furong Hall and cafeteria
8:00-8:30	Check in	Sunshine Hall
8:30-9:00	Opening ceremony	Sunshine Hall
9:00-10:20	Plenary lecture (Section I)	Sunshine Hall
10:20-10:40	Coffee break	Coffee break zone
10:40-12:00	Plenary lecture (Section II)	Sunshine Hall
12:00-13:30	Lunch (Buffet)	Furong Hall and cafeteria
13:30-14:50	Oral presentation	VIP3, VIP4, 205, 206, 211, 212, 215
14:50-15:10	Coffee break	Coffee break zone
15:10-16:30	Oral presentation	VIP3, VIP4, 205, 206, 211, 212, 215
16:30-18:00	Poster presentation	Coffee break zone
18:00-19:30	Dinner (Buffet)	Furong Hall and cafeteria
5 th Nov., 2013		
7:00-8:30	Breakfast	Furong Hall and cafeteria
8:30-10:10	Oral presentation	VIP3, VIP4, 205, 206, 211, 212, 215
10:10-10:40	Coffee break	Coffee break zone
10:40-12:00	Oral presentation	VIP3, VIP4, 205, 206, 211, 212, 215
12:00-13:30	Lunch (Buffet)	Furong Hall and cafeteria
13:30-14:30	Oral presentation	VIP3, VIP4, 205, 206, 211, 212, 215

14:30-14:50	Coffee break	Coffee break zone
14:50-16:55	Distinguished Young Scientist Lecture	Sunshine Hall
16:55-17:15	Representative Enterprise Lecture	Sunshine Hall
18:30-22:00	Farewell banquet	Sunshine Hall
6 th – 7 th Nov., 2013		
All days	Tours	

Conference venue (The second floor of the Sunshine Hotel & Resort Zhangjiajie)



Plenary lecture: Section I

Chairmen: Dingfan Qiu, Beijing General Research Institute of Mining and Metallurgy

Shuji Owada, Waseda University

- 9:00-9:40, 4th Nov.
Topic: The Policy and Technology Development for Treatment and Recycling of Heavy Metal Pollutants
Lecturer: Xiaobo Min
Professor
Central South University
- 9:40-10:20, 4th Nov.
Topic: Status and Technological Trend of Metal Recycling in Korea

Lecturer: Joonsoo Kim
Principal Researcher
Korea Institute of Geoscience and Mineral Resources

Plenary lecture: Section II

Chairmen: Juu-En Chang, National Cheng Kung University

Hyungkyu Park, Korean Institute of Resources Recycling

- 10:40-11:20, 4th Nov.
Topic: Recovery of rare earths from phosphate gypsum
Lecturer: Junji Shibata
Professor
Kansai University
- 11:20-12:00, 4th Nov.
Topic: Engineering Feasible Process Developments for Recycling
Lecturer: H. Paul Wang
Distinguished Professor
National Cheng Kung University

Oral presentation schedule

	Section A	Section B	Section C	Section D	Section E
VIP3	Topic 2	Topic 2	Topic 2	Topic 2	Topic 1&2
VIP4	Topic 1	Topic 1	Topic 1	Topic 2	Topic 2
205	Topic 3	Topic 3	Topic 3	Topic 3	Topic 3
206	Topic 4	Topic 4	Topic 2	Topic 2	Topic 3&6
211	Topic 5	Topic 5	Topic 5	Topic 2	Topic 2
212	Topic 6	Topic 6	Topic 6	Topic 6	Topic 6
215	Topic 7	Topic 7	Topic 7	Topic 7	Topic 7

Section A: 13:30-14:50, 4th Nov., 2013

Room	Time	No., title, lecturers and affiliations
Topic 1 Chairmen: Kung-Yuh Chiang, Feng-Chia University Heechan Cho, Seoul National University		
VIP4	13:30-13:50	I-02-14. Process Assessment of Recycling of Waste White Goods in Korea. Kihong Kim, Heechan Cho, Jinan Jeong, Seoul National University Sookyoung Kim, Korea Institute of Geoscience and Mineral Resource
	13:50-14:10	I-03-26. A study on Application of Geopolymeric Green Cement. Y. S. Dai, L. C. Yang, Y. C. Ding, C. W. Jhong, T. W. Cheng, National Taipei University of Technology K. L. Lin, National Ilan University

	14:10-14:30	I-05-40. Investigation of Chlorination-Volatilization Behavior of Valuable Metals in Jewelry Grinding Sludge. Kazutoshi Haga, Atsushi Shibayama, Akita University
	14:30-14:50	I-18-219. An Overview on Utilization of Coal Tar Pitch and Its Development in China. Yuhui Ma, Wei Su, Qunhui Wan, University of Science and Technology Beijing, Chunyan Shao, Xiangguo Huang, Shenyang Academy of Environmental Sciences, Jin Yuan, Shanxi Ecological Environment Research Center
Topic 2 Chairmen: Cheng-Yan Wang, Beijing General Research Institute of Mining and Metallurgy Hitoshi Ohya, the University of Kitakyushu		
VIP3	13:30-13:50	II-01-7. Flotation of Anode Slime in Copper Electro-refining Mayumi Ito, Takashi Ikemoto, Naoki Hiroyoshi, Hokkaido University
	13:50-14:10	II-20-67. A Study on Recovery of Silicon Carbide from Silicon Sawing Waste by Hydrocyclone Chih-Wei Hsu, Wei-Hao Lee, Ta-Wui Cheng, Chin-Shan Lin, National Taipei University of Technology
	14:10-14:30	II-75-215. Fundamental Process of Iron Nanoparticle Synthesis for the Recovery of Neodymium Magnet Yonghwang Ha, Ryun Ji Gang, Hea-Bin You, Jong-Gwan Ahn, Jungwon University, Ho-Sung Yoon, Korea Institute of Geoscience and Mineral Resources
	14:30-14:50	II-63-193. Recovery of Potassium Chloride from Sintering Fume of Ironmaking Works Guang Zhan, Zhancheng Guo, University of Science and Technology Beijing
Topic 3 Chairmen: Zhong-Han Zhang, Guangzhou Research Institute of Nonferrous Metals SOHN HO-SANG, Kyungpook National University		
205	13:30-13:50	III-04-35. Recovering Gold from IC Chip by Using Superheated Steam. Fumihiro Kondoh, Hitoshi Ohya, University of Kitakyushu Nobuhiro Inoue, Yuji Komori, Fine metal Div., Astec Irie Co., Ltd.
	13:50-14:10	III-16-95. Separation of Manganese from Ternary Cathodic Material of Spent Lithium Ion Battery Using Mixture of Versatic 10 acid and PC88A Dong Ju Shin, Sung-Ho Joo, Shun Myung Shin, Korea Institute of Geoscience & Mineral Resources, Korea University of Science & Technology
	14:10-14:30	III-21-108. Leaching of Y from Spent Fluorescent Powder with Nitric Acid Ching-Hwa Lee, Ching-Hua Liao, Chi-En Hung, Elon-Ishmael Cadogan,

		Ying-Wen Chan, Fan-Yuan Chang, Chia-Hao Kuo, Huang-Chi Chen, Da-Yeh University
	14:30-14:50	III-27-221. Measuring the Potential of E-waste Recycling. Xianlai Zeng, Jinhui Li, Tsinghua University
Topic 4 Chairmen: Churl Kyoung Lee, Kumoh National Institute of Technology Hui Liu, Central South University		
206	13:30-13:50	IV-4-74. Experimental Study on Deeply Purifying Ink Wastewater Using Navel Orange Peel Yunnen CHEN, Dongshuang WANG, Chunhua LIU, Lichao DING, Jinxia NIE, Jie ZENG, Jiangxi University of Science and Technology
	13:50-14:10	IV-12-195. Synthesis of Nano-precipitated Calcium Carbonate by Carbonation Process Through Loop Reactor: Application to Waste Plastic Chul-Seoung Baek, Jin-Sang Cho, Kye-Hong Cho, Korea institute of limestone & advanced Materials, Ji-Whan Ahn, Korea Institute of Geoscience and Mineral Resources, Choon Han, Kwangwoon University
	14:10-14:30	IV-09-133. Development of Lead Nano-Powder Using Waste CRT Glass as Raw Material Fu-Shen Zhang, Mingfei Xing, Chinese Academy of Sciences
	14:30-14:50	IV -13-217. Conjunctive Effect of Topdressing and Reclaimed Municipal Wastewater Irrigation on Rhizosphere Soil Nitrogen Supplying Capacity of Greenhouse Tomato Ping Li, Chao Hu, Xuebin Qi, Zhijuan Zhao, Xiangyang Fan, Chinese Academy of Agricultural Sciences
Topic 5 Chairmen: Chao-Bo Tang, Central South University Chih-Ta Tsai, National Cheng Kung University		
211	13:30-13:50	V-09-55. The Effective Carbon Dioxide Sequestration by Mineral Carbonation of Waste Cement with Flue Gas Myoungwook Mun, Seoul National University Heechan Cho, National Cheng Kung University
	13:50-14:10	V-01-3. Study on Removal of Water-soluble Chloride from Crude Zinc Oxide and the Dechlorination of Wastewater by Synthesis of LDHs Wai-Hao Lee, Ta-Wui Cheng, National Taipei University of Technology, Teng-Ker Hsu, Chia-Cheng Wu, CHC Resource Corporation
	14:10-14:30	V -12-135. Preparation of Active Ceria and Its Desulfurization Properties Qingchun Yu, Yong Deng, Wang Fei, Baoqiang Xu, Dachun Liu, Bin Yang, Kunming University of Science and Technology
	14:30-14:50	V-15-205. Effect of Water Content on Accelerated Carbonation of MSWI Bottom Ash with CO₂ Gas and the Carbonation Kinetics

		Namil Um, Ji-Whan Ahn, Korea Institute of Geoscience and Mineral Resources Seong-Young Nam, Inha University Young-Jun Ahn, Kwangwoon University
Topic 6 Chairmen: HIROSHI NAKAZAWA, Iwate University Jun-Yi Wu, National Cheng Kung University		
212	13:30-13:50	VI-01-19. Modeling of Column Test for Metal Contaminated Soil Washing under Saturated/unsaturated Flow Taisuke Sakakibara, Kyohei Takeshita, Chiharu Tokoro, Waseda University Satoshi Kawakami, Dowa Eco-system Co.,Ltd.
	13:50-14:10	VI-03-59. Recovery of Zinc from Hydrometallurgical Zinc Residue Zhang yali , Yu xianjin, Yang jing, Yang liqun, Shijuanhua, Shandong University of Technology
	14:10-14:30	VI -04-70. Resource Recycling of Copper-contaminated Sludge Derived from Wastewater Treatment with Bayer Process Shao-Hua Hu, Sheng-Fong Sie, Dahan Institute of Tech
	14:30-14:50	VI-13-260. Removal of impurities in aurum tailings by bio-leaching and the comprehensive utilization of it Xu-Jianing, Yang-Hongying, Northeastern University
Topic 7 Chairmen: Kyung-Ho Park, Korea Institute of Geoscience and Mineral Resources Shao-Peng Li, Institute of Process Engineering, CAS		
215	13:30-13:50	VII-22-162. Research on the Process Mineralogy Characteristics of Bauxite from Guizhou Province Yueqin Qiu, Qin Zhang, Guizhou University
	13:50-14:10	VII-13-57. Characteristics of Mercury Vapor from Spent Fluorescent Lamps. Seung-Whee Rhee, Hun-Su Park, Kyonggi University
	14:10-14:30	VII-01-17. DEM Simulation for the Optimization of a Wet-type Media Agitation Grinding Mill. Kentaro Hayashi, Yuki Tsunazawa, Chiharu Tokoro, Waseda University Hideaki Iituka and Osamu Ishikawa, Nippon coke & engineering co., ltd.
	14:30-14:50	VII-18-134. Process and Kinetics of Cyanide Barren Solution by Ion Exchange Resin. YU Xian-jin, ZHANG Ya-li, FANG Tao, YANG Jing, Shandong University of Technology

Section B: 15:10-16:30, 4th Nov., 2013

Room	Time	No., title, lecturers and affiliations
Topic 1 Chairmen: Ya-Min Hsieh, Taiwan Shoufu University Kazuharu Yoshizuka, the University of Kitakyushu		
VIP4	15:10-15:30	I -16-189. Prediction of the Distribution of Minor Elements in Co-smelting of WEEE and Copper Concentrate. Youqi Fan, Nanbing Xu, Zhongshen Hua, Zhuo Zhao; Anhui University of Technology, Songwen Xiao, Changsha Research Institute of Mining & Metallurgy
	15:30-15:50	I -17-211. Fundamental Theory and Approach of Low Temperature Alkaline Smelting in Non-ferrous Metallurgy. Xueyi Guo, Jingxin Liu, Qing-hua Tian, Dong Li, Central South University
	15:50-16:10	I -01-13. Sustainable Solutions for Municipal Solid Waste Management in Bangkok. Somsak Saisinchai, Chulalongkorn University
	16:10-16:30	I -10-139. Study on the Informatization of Renewable Resource Industry. Bu Xinxin, Nankai University
Topic 2 Chairmen: Shao-Hua Hu, Dahan Institute of Technology Man Seung Lee, Mokpo National University		
VIP3	15:10-15:30	II -76-218. Dry Slag Granulation, a Sustainable Option for Comprehensive Utilization of Molten Blast Furnace Slag Yuelin Qin, Chenguang Bai, Xuewei Lv, Jie Zhang, Jinlong Hao, Chongqing University
	15:30-15:50	II -16-63. Selective Reduction of Cu from Aqueous Solutions through Hydrothermal Route for Production of Cu Powders. K.H. Park, C.W. Nam, Korea Institute of Geoscience & Mineral Resources, D. Mishra, K.K. Sahu, National Metallurgical Laboratory
	15:50-16:10	II -03-34. Magnesium Refining by Room-temperature Electrolysis in Non-aqueous Electrolyte. Jesik Park, Churl Kyoung Lee, Kumoh National Institute of Tech Kyoungjung Kwon, Sejong Univ. Hansu kim, Hanyang Univ.
	16:10-16:30	II -53-160. Recycle of Wastewater From Lead-zinc Sulfide Ore Flotation Process by Ozone/BAC Technology. Liu Xingyu, Chen Bowei, Li Wenjuan, Song, General Research Institute for Nonferrous Metals
Topic 3 Chairmen: Yi-Min Zhang, Wuhan University of Technology Keitaro Koga, JX Nippon Tsuruga Recycle., LTD		

205	15:10-15:30	III-18-97. Improvement of Sorting Technologies for the Recycling of Waste Electrical Home Appliances. Kang-In Rhee, Jeongsoo Sohn, Donghyo Yang, Sookyoung Kim, Korea Institute of Geoscience and Mineral Resources
	15:30-15:50	III-28-233. Leaching of Copper from Scrap Silicon Wafers by Hydrochloric Acid Ying-Wen Chang, Ching-Hwa Lee, Chi-En Hung, Elon Ishmael Cadogan, Ching-Hua Liao, Da-Yeh University Juu-En Chang, Wei-Sheng Chen, National Cheng Kung University
	15:50-16:10	III-03-20. DEM Simulation for Comminution Detachment Process of Waste Printed Circuit Boards Using Particle-based Rigid Body Model. Kazuki Tahara, Yuki Tsunazawa, Daiki Fujihashi, Chiharu Tokoro, Shuji Owada, Waseda University
	16:10-16:30	III-07-42. Improvement of Metal Sorption onto Chitosan Microspheres. Ching-Hwa Leea, Hang-Yi Lina, Elon Ishmael Cadogana, Chi-En Hunga, Ching-Hua Liaoa Chand-Jui Yua, Da-Yeh University Srinivasa Rao Popurib, The University of the West Indies
Topic 4 Chairmen: Hyunseok Yang, Institute for Advanced Engineering Gangfeng Liu, Waseda University		
206	15:10-15:30	IV-5-82. Development and Evaluation of the Refrigerant Extraction Needle for Hazardous Refrigerant Recovery from Waste Refrigerator Hyunseok Yang, Geonhong Kim, Mansik Kong, Institute for Advanced Engineering Kiejin Park, Ajou University Bo Saeng Kim, Association of Electronics Environment Gwangwon Lee, Weltop Technos Corporation
	15:30-15:50	IV-03-61. Basic Research on Metal Recovery from ASR by Using the Air Table. Hideaki Wada, Hitoshi Ohya, The University of Kitakyushu, Masanobu Gondoh, Kyushu Metal Industry CO,LTD.
	15:50-16:10	IV-10-159. Development of an Adaptive Design Supporting System for Dismantling Plant of End-of-Life Vehicles. Young Tae Sohn, Myon Woong Park, Korea Institute of Science and Technology
	16:10-16:30	IV-06-83. Recycling Process for Multi-Layer Packaging Film Wastes to PO Pellets. Young-Chul Lee, Hyung-Chan Lee, Myeong Jun Kim, Seok-Kyu Park, Young Jae Kim, Korea Institute of Industrial Technology
Topic 5 Chairmen: Young-Chul Lee, Korea Institute of Industrial Technology		

You-Cai Zhao, Tongji University		
211	15:10-15:30	V-14-194. Development of CCUS Technologies For Mining Waste Removal by Accelerated Carbonation Thenepalli Thriveni, Um Nam Il, Ji-Whan Ahn, Korea Institute of Geoscience and Mineral
	15:30-15:50	V-02-4. The Research on Primary Sedimentation Sludge Reduction with the Addition of Liquid Fertilizer Wai-Hao Lee, Ta-Wui Cheng, Teng-Ker Hsu, Chia-Cheng Wu, National Taipei University of Technology
	15:50-16:10	V-06-29. Utilization of Manganese Oxides in Spent Carbon-zinc Batteries as Adsorbent and Oxidant of Arsenite Hiroshi Nakazawa, Wataru Hareyama, Iwate University Satoshi Tsunematsu, ES General Laboratory Co.Ltd.
	16:10-16:30	V-05-28. The Application of Hot Rolling Sludge on CSC's Sludge-coal Fly Ash Mixture System Yi-Chen Liu, China Steel Corporation
Topic 6 Chairmen: Jae-woo Ahn, Daejin University Jun-Xue Zhao, Xi'an University of Architecture and Technology		
212	15:10-15:30	VI-02-32. Preparation of Heavy Metal Adsorbent From Heavy Oil Ash Using Sulfur-impregnation, and Its Removal Ability For Lead from Aqueous Solution. Takaaki Wajima, Chiba University
	15:30-15:50	VI-17-161. Experimental Study on Recovery of Nickel from Nickel-bearing Laterite. Xianhai Li, Biyang Tuo, Qin Zhang, Shenjun Zhang, Guizhou University
	15:50-16:10	VI-12-137. Recycle of Detoxified Municipal Solid Waste Incinerator Fly Ash for Fabricating Value-added Ceramic Tile. Kun-Sen Changa, Hsiung-Wen Hanb, Mei-Fen Liua, Ya-Ting Yanga, Tzu-Ting Wanga, Chia-Hao Lua, Chen-Wei Tana, National United University
	16:10-16:30	VI-11-132. An Evaluation of the Effectiveness of Novel Industrial By-products and Organic Wastes on Heavy Metal Immobilization of Pb/Zn Mine Tailings. Yang Sheng-Xiang, Cao Jian-Bing, Hu Wen-Yong, Zhang Xiao-Jun, Zhu Jie, Jishou University
Topic 7 Chairmen: Yen-Pin Fang, TAIWAN STEEL UNION CO., LTD. Mikiya Tanaka, National Institute of Advanced Industrial Science and Technology		
215	15:10-15:30	VII-02-21. Investigation of Intermediate Products on the Reduction of Carbene Dioxide to Methanol/Ethanol using a Cyclic Fenton Reaction. Fumiya Yamagishi, Daisuke Haraguchi, Chiharu Tokoro, Waseda University

		Toshiaki Murata, Mitsui Zosen Plant Engineering, Inc.
	15:30-15:50	VII-30-191. Study on Advanced Treatment of Tannery Wastewater by Biological Velocity Discrete System Chunhua Liu, Hong Yao, Yue Zhang, Xiaodong Sang, Changbin Xia, Changsha University of Science and Technology
	15:50-16:10	VII-23-164. A Study of Reusing Spent Chemicals from Laboratory Wei-Sheng Chen, Chang-Wen Lin, Jhong-Lin Wu, Yen-Ting Lai, National Cheng Kung University
	16:10-16:30	VII-24-172. A New Technique for Removing Chlorine and Fluorine from Intermediate Material of Lead and Zinc Metallurgy by Microwave Roasting. GUO Zhanyong, JU Shaohua, PENG Jin-hui, ZHANG Li-bo, Wei Yaqian, Kunming University of Science and Technology

Section C: 8:30-10:10, 5th Nov., 2013

Room	Time	No., title, lecturers and affiliations
Topic 1		
Chairmen: Xue-Yi Guo, Central South University Wataru Hareyama, Iwate University		
VIP4	8:30-8:50	I -07-87. Institutional Supporting Plans to Activate Resource Recirculation of Waste Rare Metals Hi Sun Lee, Eun Mo Yang, Korea Environment Institute
	8:50-9:10	I -04-33. The Evaluation of Carbon Emission and Economy of the Processes of UV/O₃ with BAC Applied in Treatment of Landfill Leachate Chiu-Yi Chan, Chi-Kang Lin, Yin-Gru Chen, TungNan University
	9:10-9:30	I -15-183. Evaluation of Meal Recycling from Printed Circuit Board by Heat Treatment and Physical Separation Toyohisa Fujita, Hiroyuki Ono, Gjergj Dodbiba, Li Pang Wang, Seiji Matsuo, Katsunori Okaya, the University of Tokyo
	9:30-9:50	I -13-155. The Practice and Exploration of Discipline Construction of Regeneration Resources Industry in China ZHU Tan, GAO Shuai, LIN Chengmiao, Nankai University
	9:50-10:10	I -14-168. Study on Production Characteristics of Distillation Residues of Typical Basic Organic Chemical Industry in China Wenlong Feng, Dexun Zou, Yanping Liu, Hairong Yuan, Baoning Zhu, Xiujin Li, Beijing city environmental pollution control and resource reuse engineering research center
Topic 2		
Chairmen: Jin-Hui Peng, Kunming University of Science and Technology Hiroyuki Matsumura, the University of Tokyo		
VIP3	8:30-8:50	II -67-204. The Sintering Behavior of Calcium Sulfoaluminate - Calcium Silicates Phases with Alumina Waste Sludge and Lime

		Stone Kwang-Suk You, Ji-Whan Ahn, Korea Institute of Geoscience and Mineral resources (KIGAM) Hyoung Woo Lee, Seung Eui Min, Keon Ho Lee, Hanil Cement Corp. Ltd
	8:50-9:10	II -23-78. Application of Steelmaking Slag as a Rehabilitation Material of Coastal Environment from Sea Desertification Hiroyuki Matsuura, Xiaorui Zhang, Likun Zang, Ming Zhong, Fumitaka Tsukihashi, The University of Tokyo
	9:10-9:30	II-10-24. Degradation of TCE Utilizing Organic Acid Included in Waste from Vitis Coignetiase Juice Industry Factory Wataru Hareyama, Atsuko Fukushi, Rebiya Maimaiti, Hiroshi Nakazawa, Iwate University
	9:30-9:50	II -46-131. Study of Waste Water Recycling for Xiaoliuguo Low Grade Poly-metallic Scheelite ore Zhang Zhonghan, Hu Hongxi, Guangzhou Research Institute of Nonferrous Metals Liu Jun, Zhou Zhengshan, Xu Fude, Gansu Xinzhou Mining Co. Ltd
	9:50-10:10	II -45-130. Application Research on Slag with Higer CaO and Lower SiO₂ in Nickel Smelting Process Junxue ZHAO, Yaru CUI, Xiaoming LI, Wendan TANG, Xi'an University of Architecture and Technology Xiaoting GAO, Jinchuan Nonferrous Metal Corporation
Topic 3 Chairmen: Bing-Cai Pan, Nanjing University Jeongsoo Sohn, Korea Institute of Geoscience and Minerals Resources		
205	8:30-8:50	III-09-69. E-waste Recycling Technologies in Korea Jinki Jeong, Jae-chun Lee, Byung-Su Kim, Dong-hyo Yang, Korea Institute of Geoscience and Mineral Resources Jun-chul Choi, Recytech Inc.
	8:50-9:10	III-06-43. Comparison of Beaker Leaching and High-temperature and High-pressure Acid Leaching of PCBs for Valuable Metals Recovery Baisui HAN, Nozomi KOTAKE, Batnasan ALTANSUKH, Kazutoshi HAGA, Akira HOSOI, Yasushi TAKASAKI, Atsushi SHIBAYAMA, Akita University
	9:10-9:30	III-11-85. Recovery of Co and Ni from Spent Lithium Ion Battery by Hydrometallurgical Process Jae-Woo Ahn, Daejin University Ki-Woong Lee, Hong-in Kim, Hyun-Tae Son, Kang-Myung Yi, Sung Eel Hitech Co., LTD. Seong-ho Son, Korea Institute of Industrial Technology
	9:30-9:50	III-24-154. Comparison Between Extraction Processes of Gold and Copper from Waste Mobile Phone PCBs

		Jianfeng Yin, Sihui Zhan, He Xu, Nankai University
	9:50-10:10	III-08-49. Underwater Explosion For Disintegrating Discarded LCDs For Recycling G. Doddiba, H. Nagai, L.P. Wang, K. Okaya, T. Fujita, The University of Tokyo K. Murata, Nippon Koki Co.Ltd
Topic 2 Chairmen: Chen-Guang Bai, Chongqing University Yeu-Juin Tuan, National Cheng Kung University		
206	8:30-8:50	II-65-199. Current Status on the Recycling of Domestic Non-Ferrous Metal Scraps Hyungkyu Park, Korea Institute of Geoscience and Mineral Resources
	8:50-9:10	II-70-209. Research on Mechanism for Recycling of Wasted Pb-Sn Alloy by Vacuum Distillation. Lingxin Kong, Bin Yang, Baoqiang Xu, Yifu Li, Dachun Liu, Yongnian Dai, Kunming University of Science and Technology
	9:10-9:30	II-18-65. The Study on the Recovery of Vanadium/Tungsten from Waste SCR Catalysts Jyh-Herng Chen, Kai-Chung Hsu, Yu-Hao Chang, National Taipei University of Technology Chi-Lang Chen, Chyi-Woei Young, China Steel Corporation
	9:30-9:50	II-39-116. Studies on Vitrification of Arsenic-containing Fly Ash from Lead Metallurgical Industry SHI Mei-qing, CHAI Li-yuan, LIANG Yan-jie, LI Mi, DONG Guo-yu, Central South University
	9:50-10:10	II-60-186. Feasibility of Wastewater Sludge to Energy by an Updraft Gasification System Kung-Yuh Chiang, Cheng-Han Lu, Shan-Shin Ton, Chih-Chao Wu, Feng-Chia University, Tai-Chung
Topic 5 Chairmen: Kun-Sen Chang, National United University Qi-Xing Zhou, Nankai University		
211	8:30-8:50	V-13-143. Researching the Processing of Preparing White Carbon Black by Using Lime Kiln Tail Gas and Low Grade Silica Fume Zhi Xin, Guo zhancheng, University of Science and Technology Beijing
	8:50-9:10	V-03-5. Modification of Expanded Bed Reactor to Improve the Efficiency of Odor Elimination of Bottom Ash Chi-Kang Lin, Chiu-Yi Chan, Long-Shih Chen, Chia-Ying Chiang, Yu-Fu Sheng, Tunghsuan University Jane-Fuh Lu, Chinese Culture University
	9:10-9:30	V-10-76. Suppression of Pyrite Oxidation by Silicate Coating with Catechol Complex Oxidation Kinetics Estimation with Electrochemistry Hajime Miki, Tsuyoshi Hirajima, Mutia Dewi Yuniati, Keiko Sasaki,

		Kyushu University
	9:30-9:50	V-07-37. Reusing Laboratory Waste Incinerator Bottom Ash as an Additive for Cement Clinker Production Ying-Liang Chen, Juu-En Chang, Wen-Jhy Lee, National Cheng Kung University Ming-Sheng Ko, National Taipei University of Technology Yi-Chieh Lai, Cheng Shiu University
	9:50-10:10	V-16-258. Recovery of Cadmium in the Aqueous Solution by an Electrochemical Deposition Approach Wenjie Xu, Qingwei Guo, Zhencheng Xu, Qiongfang Zhuo, South China Institute of Environmental Sciences
Topic 6 Chairmen: Shun Myung Shin, Korea Institute of Geoscience & Mineral Resources (KIGAM) Motoo Sato, SATO TEKKO CO.,LTD.		
212	8:30-8:50	VI . Preparation of aluminum-silicon refractory materials from high-alumina fly ash by mineralogical structure regulation Shao-Peng Li, Institute of Process Engineering, CAS
	8:50-9:10	VI-06-113. Zinc Recovery from Zinc Calcine by Selective Reduction Roasting Dongke Hou, Bing Peng, Ning Peng, Huan Yan, Ming Hu, Central South University
	9:10-9:30	VI-20-176. Investigation on Endurance and Adsorption Behavior of Miscanthus to Pb and Zn in Soil Luo Lin, Mao Shihua, Wei Jianhong, Peng Xiangqi, Hunan Agricultural University
	9:30-9:50	VI -18-166. Polymer-based Nano-Zirconium Oxide for Highly Efficient Removal of Toxic Metals from Water Yingnan Jiang, Ming Hua, Bingcai Pan, Quanxing Zhang, Nanjing University
	9:50-10:10	VI-21-177. Simultaneous Recovery of Zinc Containing EAF Dust and Treatment of Contaminated Soil by Waelz Kiln Technology Kenneth Fang, Taiwan Steel Union, Corp Mark Lin, the Formosa Association of Resource Recycling
Topic 7 Chairmen: Hi Sun Lee, Korea Environment Institute Ching-Hwa Lee, Da-Yeh University		
215	8:30-8:50	VII-19-144. Effects of Surfactant Addition on Waste Activated Sludge Hydrolysis and Short-chain Fatty Acids Accumulation Can Chen, Cheng Huang, Yubo Yan, Na Li, Junping Wen, Fangbian Ma, Xiuyun Sun, Nanjing University of Science and Technology
	8:50-9:10	VII-08-46. Kinetic Evaluation of Surface Grinding by Attoritor for the Concentration of PGMs from Spent Automobile Catalysts Tomoki ICHINOSE, Ganfeng LIU, Ayumu TOKUMARU, Shuji OWADA, Waseda University

	9:10-9:30	VII-20-153. Behavior and Removal of Impurities in Alkaline Zinc Electrolyte from Waste Products Qiang Li, Youcai Zhao, Tongji University
	9:30-9:50	VII-25-174. Water use Efficiency and Potato Produce Using Low Quality Waters by Water-saving Irrigation Technology Qi Xue-Bin, Qiao Dong-Mei, Li Ping, Huang Zong-Dong, Chinese Academy of Agricultural Sciences
	9:50-10:10	VII-09-47. Selection of Effective Filter Material for the Precise Detection of Non-Ferrous Materials in XRT Sorting Yohei Goto, Shuji Owada, Waseda University

Section D: 10:40-12:00, 5th Nov., 2013

Room	Time	No., title, lecturers and affiliations
Topic 2 Chairmen: Bing Peng, Central South University Yasushi Takasaki, AKITA University		
VIP4	10:40-11:00	II -55-170. Application of Xanthates to Recover Cu/Mo/Zn in Optoelectronic industry Wastewater Hsing-Cheng Lu, Juu-En Chang , National Cheng Kung University Yi-Kuo Chang, Central Taiwan University of Science and Technology Ramadian Pravitasari, National Cheng Kung University
	11:00-11:20	II -27-86. A Study on the Physical and Chemical Properties and Resource Technology of Sediment of Waste Cutting Oil Ya-Min Hsieh, Wann-Lee Liou, Taiwan Soufu University Mei-Fang Wu, Wei-sheng Chen, National Cheng Kung University
	11:20-11:40	II -44-126. Deep Cleaning of Copper Slag from an Electric Cleaning Furnace Wei-jiao Yang, Cheng-yan Wang, Bao-zhong Ma, Fei Yin, Yong-qiang Chen, Beijing General Research Institute of Mining and Metallurgy
	11:40-12:00	II -36-110. Recovery of Nitric Acid and Copper from Plating Waste of Automobile Wheel Yonghwang Ha, Ryun Ji Gang, Jong-Gwan Ahn, Jungwon University Seong Ho Son, Wonsik Lee, Korea Institute of Industrial Technology,
Topic 2 Chairmen: Junji Shibata, Kansai University Xiao-Bo Min, Central South University		
VIP3	10:40-11:00	II -13-54. An Analysis of the Characteristics of Recovered High-purity Silicon Carbide Powder Hui-yi Tsai, Chun-yi Wu, Wei-sheng Chen, Tuu-en Chang, Kuan-yu Shih, National Cheng Kung University
	11:00-11:20	II-51-152. Separation of Chromium from Sulphuric Acid Leaching Solutions of Mixed Electroplating Sludge by Selective Phosphate Precipitation

		Zhifeng Xu, Feng Guo, Rihui Yue, Jiangxi University of Science and Technology Chengyan Wang, Wenhui Yuan, Bo Yang, Beijing General Research Institute of Mining and Metallurgy
	11:20-11:40	II -48-141. Study on the Making of Porous Sound Absorbing Material Using Waste Steel Slag Peng Li, Zhancheng Guo, University of Science and Technology Beijing
	11:40-12:00	II -54-167. Experiment on Preparation of Fly Ash I by Flotation of Decarbonization Xiaodong Cheng, Xiaotai He, Song Mao, Qin Zhang, Guizhou University
Topic 3 Chairmen: Hong-Ying Yang, Northeastern University Haiying Wang, Central South University		
205	10:40-11:00	III -02-11. Research on the Flotation Separation of Electronic Ceramic Powder and Glass Fiber Resin of Electronic Waste Xiaoping Xu, Yong Liu, Xianyang Qiu, Guangzhou Research Institute of Nonferrous Metals/ Guangdong General Research Institute of Industrial Technology
	11:00-11:20	III-25-165. Synthesis and Performances of Spherical $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ Prepared from Spent Lithium-ion Batteries Yue Yang, Yaqing Weng, Xuejun Wang, Guoyong Huang and Shengming Xu, Tsinghua University
	11:20-11:40	III -13-89. Reductive Ammonia Leaching of Spent Lithium-ion Battery Cathode Materials by Ammonium Sulfite Jeongsoo Sohn, Sookyoung Kim, Donghyo Yang, Kangin Rhee, Korea Institute of Geoscience & Mineral Resources(KIGAM) Heesuk Ku, Kyungjung Kwon, Sejong University
	11:40-12:00	III -12-88. Development of Image Processing System for Air Classification Atsunori Tayaoka, Eriko Tayaoka, Kitakyushu National College of Technology Tsuyoshi Hirajima and Keiko Sasaki, Kyushu University
Topic 2 Chairmen: Fu-Shen Zhang, Chinese Academy of Sciences Tasuma Suzuki, Yamaguchi University		
206	10:40-11:00	II -82-242. The Study of Flootation Separation for Silicon Slurry Chun-Chi Li, Kuan-Yu Shih, Yun-Hwei Shen, National Cheng Kung University Ya-Min Hsieh, Taiwan Shoufu University Wei-Sheng Chen, National Cheng Kung University
	11:00-11:20	II -21-72. Recycling of Silicon Sludge to Silicon Tetrachloride and Porous Carbon via Chlorination Kyun Young Park, Hoey Kyung Park, Bong Whan Ko, Mi Sun Lee, Tae

		Won Kang, Kongju National University Hee Dong Jang, Korea Institute of Geoscience and Mineral Resources (KIGAM)
	11:20-11:40	II-29-91. Research on the New Technology of Preparation of High Purity Iron (III) Oxide from the Zinc Hydrometallurgy High Iron Slag by Hydrothermal Method Ming-yan Lan, Jing He, Jun-le Lu, Rui Guo, Jia-wei Zhang, Central South University
	11:40-12:00	II-34-103. Study on the Preparing Co₃O₄ from Blister Refining Cobalt-rich Slag by Ozonation Qinghua Tian, Yuntao Xin, Xueyi Guo, Tinggang Yuan, Hengli Wang, Central South University
Topic 2 Chairmen: Jin-Young Lee, Korea Institute of Geoscience and Mineral Resources Jyh-Herng Chen, National Taipei University of Technology		
211	10:40-11:00	II-81-237. Community Structure and Dynamics during Bioleaching of Low Grade Copper Sulfide at Different pHs Yuguang Wang, Weimin Zeng, Guanzhou Qiu, Lili Wan, Kai Li, Hongbo Zhou, Central South University
	11:00-11:20	II-28-90. Utilization of Smelting Slag in Eco-friendly Metallurgical Process of Cobalt Concentrate Jeongsoo Sohn, Sookyung Kim, Donghyo Yang, Kangin Rhee, Cheolwoo Nam, Korea Institute of Geoscience & Mineral Resources(KIGAM)
	11:20-11:40	II-50-147. Mineralogical Analysis and Hydraulic Cyclone Separation on the Carbide Slag Shibin Yang, Rongyi Lin, Huiquan Li, Guohua Li, China university of mining and technology(Beijing)
	11:40-12:00	II-24-80. Application of Bottom Ash from Coal Power Plants to the Polymer Composite Materials Ha Youn Kim, Jae Won Choi, Yong-Chan Chung, Byoung Chul Chun, Inje University
Topic 6 Chairmen: Bin Yang, Kunming University of Science and Technology Mayumi Ito, Hokkaido University		
212	10:40-11:00	VI-24-200. Column Experiment on Cr(VI) Reduction with SiO₂-coated Fe Nanocomposites Yongchao Li, Kejia Liu, Bozhi Ren, Hunan University of Science and Technology
	11:00-11:20	VI-16-151. Application of Fungus for Phytoextraction of Heavy Metal-contaminated Soil Shengcui Zhu, Xiaoxi Zeng, Jianxin Tang, Benjie Wei, Hunan University of Technology
	11:20-11:40	VI-26-220. Removal of Arsenic by Sodium Sulfide Precipitation and

		Recovery of Iron by Hematite Process from Zinc Sulphate Solution Jing He, Mingyan Lan, Jiawei Zhang, Fuqi Zhang, Central South University
	11:40-12:00	VI-27-224. Separation and Recycle of Heavy Metals from Wastewater by Synergistic Solvent Extraction Method: A Review Yinxuan Qiu, Limei Yang, Songtao Huang, Zheng Xu, General research institute for nonferrous metals
Topic 7 Chairmen: Ji-Whan Ahn, Korea Institute of Geoscience and Mineral Resources(KIGAM) Hsing-Cheng Lu, National Cheng Kung University		
215	10:40-11:00	VII-28-182. Research and Application of Synergistic for Beneficiation Guofang Du, Fei Xiong, Yaozhong Lan, Yunnan University
	11:00-11:20	VII-21-158. Adsorption of Methylene Blue from Aqueous Solution by Cement Granular Red Mud Le. T. Q. Xuan, Shaohua Ju, Liexing Zhou, Peng Jin-hui, Xia Yi, Yunnan provincial Key Laboratory of Intensification Metallurgy
	11:20-11:40	VII-07-45. Concentration of Minor Rare Metals from E-scrap by Combining New Comminution and Physical Separation Processes Shuji Owada, Faculty of Science and Engineering, Waseda University Takamasa Komuro, Tatyuki Ono, Rintaro Hayashi, Waseda University
	11:40-12:00	VII-32-198. Degradation of Rhodamine B by Ultrasound/Fenton-Like Reagent Bozhi Ren, Hongpu Ma, Zhang Peng, Zheng Xie, Hunan University of Science and Technology

Section E: 13:30-14:30, 5th Nov., 2013

Room	Time	No., title, lecturers and affiliations
Topic 1 and 2 Chairmen: Kyehong Cho, Korea Institute of Limestone and Advanced Materials Katsunori Okaya, University of Tokyo		
VIP3	13:30-13:50	I -09-128. Assessment of Heavy Metal Pollution in Topsoil of Nonferrous Metals Mining Area Shun-hong Huang, Wei Chen, Cui-yu Yuan, Yi Xu, Hunan Research Institute of Nonferrous Metals Zhen-xing Wang, South China Institute of Environmental Sciences
	13:50-14:10	II-14-56. Recycling of Waste Silicon Sludge Using Separation and Purification Processes in Taiwan Jun-Yi Wu, Wei-Sheng Chen, Min-Shing Tsa, Juu-En Chang, National Cheng Kung University
	14:10-14:30	II-56-171. Current Status and Recovery of Valuable Materials for WEEE in Taiwan Jun-Yi Wu, Wei-Sheng Chen, Min-Shing Tsa, Juu-En Chang, National Cheng Kung University
Topic 2		

Chairmen: Kwang-Suk You, Korea Institute of Geoscience and Mineral resources (KIGAM) Somsak Saisinchai, Chulalongkorn University		
VIP4	13:30-13:50	II -15-60. Electrochemical Recovery of Gallium and Arsenic from Wastes in Deep Eutectic Solvents Chung-Jui Su, H. Paul Wang, Fang-Chih Chang, Chia-Hsu Shen, I-Wen Sun, National Cheng-Kung University
	13:50-14:10	II -61-188. Preparation of Fe/Al composite for metal recovery Mayumi Ito, Kota Sukegawa, Naoki Hiroyoshi, Hokkaido University
	14:10-14:30	II -64-196. Characteristics of Natural Hydraulic Lime-based Mortar in Korea Jinsang Cho, Kiyeon Moon, Moonkwan Choi, Kyehong Cho, Korea Institute of Limestone and Advanced Materials Jiwhan Ahn, Korea Institute of Geoscience and Mineral Resources
Topic 3 Chairmen: Haga Kazutoshi, AKITA University Jian-Guang Yang, Central South University		
205	13:30-13:50	III-20-106. Development of an Apparatus to Disassemble Electronic Components from Printed Circuit Board Seungsoo Park, Seongmin Kim, Insang Jung, Yosep Han, Joonchul Choi, JaiKoo Park, Hanyang University
	13:50-14:10	III-10-75. Sulfur-fixing Reduction Smelting Spent Lead-acid Battery Colloid Sludge in Fused Sodium Salt at Low Temperature YU-JIE HU, CHAO-BO TANG, MO-TANG TANG, YONG-MING CHEN, SHENG-HAI YANG, JIAN-GUANG YANG, and JING HE, Central South University
	14:10-14:30	III -17-96. “Cross Flow Shredder”: Novel Tool for Concentrating Minor Rare Metals from E-scrap Moto-o Sato, Katsushi Okumura, Hiroshi Yamada, Naoki Sunahara, Hideaki Fukusawa, Sato Tekko Co., Ltd. Shuji Owada, Chiharu Tokoro, Faculty of Science and Engineering, Waseda university
Topic 3 and 6 Chairmen: Haijime Miki, Kyushu University Xing-Yu Liu, General Research Institute for Nonferrous Metals		
206	13:30-13:50	III -15-94. Preparation of Co-Br-Mn Catalyst from Lithium Ion Batteries Scrap by Solvent Extraction with Na-Cyanex 272. Dong Ju Shin, Sung-Ho Joo, Shun Myung Shin, Korea Institute of Geoscience & Mineral Resources
	13:50-14:10	III19-105. Adsorption Characteristics of Co(II) and Mn(II) on the Functionalized Mesoporous Silica. SeongMin Kim, SeungSoo Park, InSang Jung, SeokJe Kwon, Aram Seo, JoonChul Choi, JaiKoo Park, Hanyang University, YoSep Han, HyunJung Kim, Chonbuk National University

	14:10-14:30	VI-25-206. Effect of Accelerated Carbonation on the Leaching of Heavy metals in Mine Tailing Seong-Young Nam, Jin Kim, Inha University Namil Um, Ji-Whan Ahn, Korea Institute of Geoscience and Mineral Resources
Topic 2 Chairmen: Atsushi Iizuka, Tohoku University Baoqiang Xu, Kunming University of Science and Technology		
211	13:30-13:50	II-58-181. Surface Grinding for Recycling of Filter Sand Used in Treatment of Acid Mine Drainage Shintaro Okada, Kentaro Hayashi, Yuki Tsunazawa, Chiharu Tokoro and Shuji Owada, Waseda University
	13:50-14:10	II-30-92. Development of Recycling Technology in Kitakyushu Using Industry-Academy Cooperation Hitoshi Ohya, Fumihiro Kondoh, Hideaki Wada, The University of Kitakyushu Nobuhiro Inoue, Yuji Komori, Astec Irie Co., Ltd. Masanobu Gondoh, Kyushu Metal Industry Co., Ltd.
	14:10-14:30	II-32-101. Fundamental and Approach of Low Temperature Alkaline Smelting in Non-ferrous Metallurgy Xueyi Guo, Jingxin Liu, Qing-hua Tian, Dong Li, Central South University
Topic 6 Chairmen: Bao-Ping Jia, Changzhou University Namil Um, Inha University		
212	13:30-13:50	VI-10-123. The Selective Reduction Roasting of High Iron-bearing Zinc Calcine Ning Peng, Bing Peng, Li-Yuan Chai, Mi Li, Huan Yan, Dong-Ke Hou, Ming Hu, Central South University
	13:50-14:10	VI-14-124. Separation of Molybdenum and Vanadium Using Two Stage Process of Solvent Extraction and Stripping Precipitation Tasuma Suzuki, Akiko Fuji, Nanae Matsunaga, Masakazu Niinae, Yamaguchi University Junji Shibata, Energy and Environmental Engineering, Kansai University
	14:10-14:30	VI-23-197. The Removal of Heavy Metals and Upgrading Crude Bio-oil from Pteris Vittata Stems and Leaves Harvest Using Hydrothermal Upgrading Process Thenepalli Thriveni, Um Nam Il, Ji-Whan Ahn, Korea Institute of Geoscience and Mineral Resources(KIGAM)
Topic 7 Chairmen: Xian-Jin Yu, Shandong University of Technology Jong-Gwan Ahn, Jungwon University		
215	13:30-13:50	VII-12-227. Copper-Indium Alloys Sputtering Waste Separation and Recovery Technology

		Shu-mei Huang, Wei-sheng Chen, Kai-Lun Chiu, Tuu-en Chang, National Cheng Kung University
	13:50-14:10	VII-11-52. Solvent Extraction Separation of Gallium and Copper from Sputtering Target Shih-Lin Huang, Wei-Sheng Chen, Jun-Yi Wu, Hsing-Cheng Lu, Kai-Lun Chiu, National Cheng Kung University
	14:10-14:30	VII-10-51. Flow Channel Applied in Microbial Fuel Cell. Chin-tsan Wang, National I-Lan University Zhao-qin Qi, Shi-jie Fan, Yantai University

Poster presentation 16:20-18:00 4th Nov., 2013

	No., title, lecturers and affiliations
1	III-01-1. Effects of Foam Agent on Characteristics of Thin Film Transistor Liquid Crystal Display (TFT-LCD) Waste Glass- Metakaolin-based Cellular Geopolymer Kae-Long Lin, Hau-Shing Shiu, National Ilan University Kang-Wei Luo, National Taipei University of Technology Ta-Wui Cheng, Chao-Lung Hwang, National Taiwan University of Science and Technology
2	V-02-4. The Research on Primary Sedimentation Sludge Reduction with the Addition of Liquid Fertilizer Chia-Ji Teng, Chi-Kang Lin, Yen-Hsiang Peng, Chi Hsiao, Yu-Syuan Sie, Tunghai University
3	V-03-5. Modification of Expanded Bed Reactor to Improve the Efficiency of Odor Elimination of Bottom Ash Chi-Kang Lin, Chiu -Yi Chan, Long-Shih Chen, Chia-Ying Chiang, Yu-Fu Sheng, Tunghai University Jane-Fuh Lu, Chinese Culture University
4	II-04-10. Effect of Incinerator Fly Ash Addition on LWA Formation Yu-Ling Wei, Jing-Chiang Yang, Tunghai University
5	II-07-16. Using Wet Chemical Synthesis with Dispersant to Prepare Zero-valent Metals and Multi Metals for Removing Sodium Pentachlorophenate in water Min-Hsin Liu, Bo-Yan Chen, Yu-Ming Chien and Kuan-Hsien Li, Chaoyang University of Technology
6	V-04-22. Separation and Recovery of Boron from Water Using Ion Exchange Methods Shuming Shao, Syouhei Nishihama, and Kazuharu Yoshizuka, the University of Kitakyushu
7	II-10-24. Degradation of TCE Utilizing Organic Acid Included in Waste from Vitis Coignetiae Juice Industry Factory Wataru Hareyama, Atsuko Fukushi, Rebiya Maimaiti, Hiroshi Nakazawa, Iwate University
8	II-12-30. Enrichment of Phosphate in Sewage Sludge Ash by Dry Magnetic Separation and Gravity Separation Kenta Odajima, Wataru Hareyama, Hiroshi Nakazawa, Iwate University Teruhito Sasaki, Iwate Industrial Research Institute
9	II-02-31. Distribution Behavior of Bi and Pb Between Molten PbO-SiO₂ Slag and Bi Bo-Ram Han, Se-Jong Kim and Ho-Sang Sohn, Kyungpook National University

10	II -03-34. Magnesium Refining by Room-temperature Electrolysis in Non-aqueous Electrolyte Jesik Park, Churl Kyoung Lee, Kumoh National Institute of Tech. Kyoungjung Kwon, Sejong Univ. Hansu kim, Hanyang Univ.
11	III-04-35. Recovering Gold from IC Chip by using Superheated Steam Fumihiro Kondoh, Hitoshi Ohya, the University of Kitakyushu Nobuhiro Inoue, Yuji Komori, Astec Irie CO., LTD.
12	II -43-251. Fundamental Process of Iron Acetate Synthesis For the Recovery of Neodymium Magnet Yonghwang Ha, Ryun Ji Gang, Hea-Bin You, Ho-Sung Yoon, Jong-Gwan Ahn, Korea Institute of Geoscience and Mineral Resources
13	II -22-73. Operation of Column Extractor for the Recovery of Pt from the Chloride Leaching Solution of Spent Catalysts Pan Pan Sun, Man Seung Lee, Mokpo National University Seong Ho Sohn, Incheon Technology Service Center
14	IV-1-44. Liquefaction of Industrial Rubber Wastes in a 24-TPD Pilot Plant F.-C. Chang, Mou-Yung Yeh, S.-F. Liu, H. P. Wang, National Cheng Kung University Y.-L. Wei, Tunghai University C.-J. G. Jou, National Kaohsiung First University of Science and Technology
15	IV-3-61. Basic Research on Metal Recovery from ASR by Using the Air Table Hideaki Wada, Hitoshi Ohya, the University of Kitakyushu Masanobu Gondoh, Kyushu Metal Industry CO., LTD.
16	VII -14-62. Fabrication of Ti-6Al-4V Powder with Low Oxygen Contents by Hydrogenation-dehydrogenation Method from Ti-6Al-4V Scraps Ki-Min Roh, Chang-Yul Suh, Wonbaek Kim, Hanjung Kwon, Jung-Min Oh, Jae-Won Lim, Korea Institute of Geoscience and Mineral Resources
17	II -17-64. Feasibility of Recycling Stainless Steel Reductive Slag as Coarse Aggregate through Cold-pressing Technique Chih-Ta Tsai, Juu-En Chang, National Cheng Kung University Chien-Chih Chang, MOTC
18	II -18-65. The Study on the Recovery of Vanadium/Tungsten from Waste SCR Catalysts Jyh-Herng Chen, Kai-Chung Hsu, Yu-Hao Chang, National Taipei University of Technology Chi-Lang Chen, Chyi-Woei Young, China Steel Corporation
19	VI -04-70. Resource Recycling of Copper-contaminated Sludge Derived from Wastewater Treatment with Bayer Process Shao-Hua Hu, Sheng-Fong Sie, Dahan Institute of Tech.
20	V -11-77. Reaction of Calcium Hydroxide with Hydrogen Halide under Simulated Bag Filter Conditions Atsushi Iizuka, Naomi Onodera, Kouki Kasuya, Etsuro Shibata, and Takashi Nakamura, Tohoku University
21	IV -5-82. Development and Evaluation of the Refrigerant Extraction Needle for Hazardous Refrigerant Recovery from Waste Refrigerator Hyunseok Yang, Geonhong Kim, Mansik Kong, Institute for Advanced Engineering

	Kiejin Park, Ajou University Bo Saeng Kim, Association of Electronics Environment Gwangwon Lee, Weltop Technos Corporation
22	IV-6-83. Recycling Process for Multi-Layer Packaging Film Wastes to PO pellets Young-Chul Lee, Hyung-Chan Lee, Myeong Jun Kim, Seok-Kyu Park, Young Jae Kim, Korea Institute of Industrial Technology
23	II -26-84. Microbial Extraction of Co and Cu from Cobalt Concentrates Using Aspergillus niger Hyo-Jin Ahn, Jae-Woo Ahn, Duk-Ki Bang, Meong-Woon Kim, Daejin University
24	III -11-85. Recovery of Co and Ni from Spent Lithium ion Battery by Hydrometallurgical Process Jae-Woo Ahn, Daejin University Ki-Woong Lee, Hong-in Kim, Hyun-Tae Son, Kang-Myung Yi, Sung Eel Hitech Co., LTD. Sung-ho Son, Korea Institute of Industrial Technology
25	VII-29-187. Water washing effects on enhanced carbonation reaction of municipal solid waste incinerator bottom ash Kung-Yuh Chiang, Cheng-Hsiang Shih, Cheng-Han Lu, Feng-Chia University
26	III-12-88. Development of Image Processing System for Air Classification Atsunori Tayaoka, Eriko Tayaoka, Kitakyushu National College of Technology Tsuyoshi Hirajima, Keiko Sasaki, Kyushu University
27	III-14-93. Produce of Zn•Mn Sulphate from Mixed Spent Batteries Shun Myung Shin, Sung-Ho Joo, Korea Institute of Geoscience & Mineral Resources (KIGAM), Korea University of Science & Technology (UST)
28	II -31-98. Demonstration Study of High Efficiency Electrolyte for Gold Recovery from Waste Solution Containing Low Concentration Gold Sookyoung Kim, Jeongsoo Sohn, Donghyo Yang, Kang-In Rhee, Korea Institute of Geoscience and Mineral Resources Youngmin Oh, Danam-ene co. LTD.
29	II -36-110. Recovery of Nitric Acid and Copper from Plating Waste of Automobile Wheel Yonghwang Ha, Ryun Ji Gang, Seong Ho Son, Wonsik Lee, Jong-Gwan Ahn, Korea Institute of Industrial Technology
30	II -41-120. Solvent Extraction of Cobalt from Co-Ni Sulfate solution using Mixer-Settler Chul-Joo Kim, Ho-Sung Yoon, Hyun-Ho Kim, Chul-Woo Nam, Korea Institute of Geoscience and Mineral Resources
31	II -42-121. Solvent Extraction of Copper from Cu-Co-Ni Sulfate Solution Using Mixer-Settler Ho-Sung Yoon, Chul-Joo Kim, Hyun-Ho Kim, Chul-Woo Nam, Korea Institute of Geoscience and Mineral Resources
32	VII -16-125. Solvent Extraction and Stripping of Boron from Brine Using 2,2,4-trimethyl-1,3-pentanediol Jyothi Rajesh Kumar, Chul-Joo Kim, Ho-Sung Yoon, Jin-Young Lee, Korea Institute of Geoscience and Mineral Resources (KIGAM)
33	III-22-122. Extraction Process of Rare Earth Elements from NdFeB Powders via

	Alkaline Treatment Kyeong Woo Chung, Chul-Joo Kim, Ho-sung Yoon, Korea Institute of Geoscience and Mineral Resources
34	II -52-157. A New Technology of Sn-Pb-Sb Alloy separation Yifu Li, Bin Yang, Baoqiang Xu, Yongnian Dai ¹ , Dachun Liu, Lingxin Kong, Guobin Jia, Kunming University of Science and Technology, Kunming
35	I -14-168. Study on Production Characteristics of Distillation Residues of Typical Basic Organic Chemical Industry in China Wenlong Feng, Dexun Zou, Yanping Liu, Hairong Yuan, Baoning Zhu, Xiujin Li, Beijing city environmental pollution control and resource reuse engineering research center
36	II -55-170. Application of xanthates to Recover Cu/Mo/Zn in Optoelectronic Industry Wastewater Hsing-Cheng Lu, Juu-En Chang, Ramadian Pravitasari, National Cheng Kung University Yi-Kuo Chang, Central Taiwan University of Science and Technology
37	II -61-188. Preparation of Fe/Al composite for Metal Recovery Mayumi Ito, Kota Sukegawa, Naoki Hiroyoshi, Hokkaido University
38	II -75-215. Fundamental Process of Iron Nanoparticle Synthesis for the Recovery of Neodymium Magnet Yonghwang Ha, Ryun Ji Gang, Hea-Bin You, Ho-Sung Yoon, Korea Institute of Geoscience and Mineral Resources Jong-Gwan Ahn, Jungwon University
39	III-27-221. Measuring the Potential of E-waste Recycling Xianlai Zeng, Jinhui Li, Tsinghua University
40	IV-14-222. Effects of Foam Agent on Characteristics of Thin Film Transistor Liquid Crystal Display (TFT-LCD) Waste Glass- Metakaolin-Based Cellular Geopolymer Kae-Long Lin, Hau-Shing Shiu, Kang-Wei Luo, National Ilan University Ta-Wui Cheng, National Taipei University of Technology Chao-Lung Hwang, National Taiwan University of Science and Technology
41	II -77-226. Recovery of Metallic Nickel from Fe-36Ni Alloy Scrap Using Magnesium Reduction Jei-Pil Wang, Pukyong National University Shun-Myung Shin, Korea Institute of Geoscience and Mineral Resources Sang-An Ha, Silla University Dong-Won Lee, Korea Institute of Materials Science Jung-Yeul Yun, Korea Institute of Materials Science
42	II -42-121. Solvent Extraction of Copper in Cu-Co-Ni Sulfate solution using Mixer-Settler Ho-Sung Yoon, Chul-Joo Kim, Hyun-Ho Kim, Chul-Woo Nam, Korea Institute of Geoscience and Mineral Resources

Distinguished Young Scientist Lectures (14:50-16:55, 5th Nov., 2013)

Chairmen: Wei-sheng Chen, the Formosa Association of Resource Recycling

Dawan Wiwattanadate, Chulalongkorn University

- 14:50-15:15, 5th Nov.

Topic: Policies and Technologies of “City Mineral” Utilization in China

Lecturer: Hui Liu
Distinguished research fellow
Central South University

- 15:15-15:40, 5th Nov.
Topic: Recovery of Metallic Nickel from Fe-36Ni Alloy Scrap using Magnesium Reduction
Lecturer: Jei-Pil Wang
Assistant Professor
Pukyong National University
- 15:40-16:05, 5th Nov.
Topic: Improvement of the Durability of Solvent Impregnated Resin Using Acidic Organophosphorus Extractant
Lecturer: Mikiya Tanaka
Ph. D.
National Institute for Pollution and Resources, Japan
- 16:05-16:30, 5th Nov.
Topic: Recycling technology development of oil-fired fly ash in Taiwan
Lecturer: Ya-Min Hsieh
Assistant Professor
Taiwan Shoufu University
- 16:30-16:55, 5th Nov.
Topic: CO₂ Capture, Storage and Utilization
Lecturer: Kreangkrai Maneeintr
Ph. D.
Chulalongkorn University

Representative Enterprise Lecture (16:55-17:15, 5th Nov., 2013)

- **The Novel development model of “City Mineral”**
New World Environmental Services Group
- **The Future of Recycling Industry**
Hunan Vary Tech Co., Ltd

Travel

Trip 1: One day deluxe tour to Yellow Dragon Cave and Baofeng Lake

Quotation: 266 RMB

After breakfast, you will visit the Yellow Dragon Cave which is known as the Best Cave in China. After lunch, visit the World Wonderland-Baofeng Lake, which is the representative spot of

Wulingyuan scenery.

Quotation inclusion: 1) All entrance fees to the scenic spot specified above; 2) Local excellent tour guide for the whole trip; 3) One lunch (Chinese Tujia flavor food, ten people a table); 4) Air conditioned tourist bus; 5) Liability insurance of the travel agency.

Quotation exclusion: 1) All cable car fees; 2) Extra costs caused by manpower irresistible factors (such as flood, landslide, rainfall, airline flight delay and cancel).

Trip 2: Two days deluxe tour to Zhangjiajie landscape and Yellow Dragon Cave

Quotation: 558 RMB

DAY 1: After breakfast, you will set off for Yuanjiajie scenic area and visit the "Hallelujah Mountain". After going down the mountain you will visit the Ten-mile Gallery.

DAY 2: After breakfast, you will visit the Yellow Dragon Cave. After lunch, enter into the first national geological forest park, Zhangjiajie National Forest Park. You will have a wonderful walk in the Golden Whip Stream scenic area, named the Most Beautiful Grand Canyon in China. At last, visit the Huangshi village scenic area.

Quotation inclusion: 1) All entrance fees to the scenic spot specified above; 2) Local excellent tour guide for the whole trip; 3) 3 meals (including two lunch and one dinner, 12 dishes per meal, ten people a table); 4) Air conditioned tourist bus; 5) Liability insurance of the travel agency.

Quotation exclusion: 1) All cable car fees; 2) Extra costs caused by manpower irresistible factors (such as flood, landslide, rainfall, airline flight delay and cancel); 3) Entry fees for Yuanjia village.

Trip 3: One day deluxe tour to Tianmen Mountain, Yellow Dragon Cave and Baofeng Lake

Quotation: 544 RMB

After breakfast, you will enter into the Tianmen Mountain National Forest Park. After a Chinese Tujia flavor lunch, you will visit the Yellow Dragon Cave. At last, visit the World Wonderland-Baofeng Lake.

Quotation inclusion: 1) All entrance fees to the scenic spot specified above, all cable car fees are included; 2) Local excellent tour guide for the whole trip; 3) One lunch (Chinese Tujia flavor food, ten people a table); 4) Air conditioned tourist bus; 5) Liability insurance of the travel agency

Quotation exclusion: Extra costs caused by manpower irresistible factors (such as flood, landslide, rainfall, airline flight delay and cancel)

Other Tips: If you are not satisfied with the above trips, please contact the travel agent in the hotel for your interested option. All the hotels in Hunan province do not provide disposable items in terms of environment protection. Visitors please take their own.

Oral presentation

Oral presentations were grouped by topics and lectured in meeting room VIP3, VIP4, 205, 206, 211, 212 and 215, respectively, in the second floor of the Sunshine Hotel & Resort Zhangjiajie. Each oral presentation composes of 15minutes for presentation and 5minutes for Q&A. Please name your PPTs following the rule: section # + topic # + your paper ID, and send them to secretariat of EARTH2013 at least half day before your presentation by email or flash disk. Lecturers are required to be present at the conference room at least 10 minutes in advance.

Poster presentation

The poster should be provided by the authors, including authors, affiliation, abstract, introduction, methods, figures, tables and conclusions. The size of the poster should be 100cm×80cm and the layout should be vertical. The poster session will be held in coffee break zone of second floor at the Sunshine Hotel & Resort Zhangjiajie. Please assemble your posters 30 minutes in advance. Presenters are asked to be present at their posters in 16:20-18:00, 4th Nov., 2013 and present your poster to the passing audiences. Each author is responsible for assembly and removal of her/his own presentation.

Registration:

Registration method:

You can download the blank registration form from the official website (<http://www.earth2013.org/download.asp>) and return the complete registration form to the secretariat of EARTH2013 by E-mail after registration online.

Registration Fees

For Chinese mainland participants (RMB)

Category	Register before 31 st Aug., 2013	Register after 1 st Sep., 2013
Regular participant	1300	1500
Student	800	1000
Accompanying person	800	1000

For Taiwan and international participants (USD)

Category	Register before 31 st Aug., 2013	Register after 1 st Sep., 2013
Regular participant	350	400
Student	200	250
Accompanying person	200	250

Regular participant and student fees cover: Plenary/Oral/Poster Session, Welcome Party, Farewell banquet, buffet, tea break, Proceedings (one copy only) and souvenir

Accompanying person fee includes: Welcome Party, Farewell banquet and buffet

Payment method:

- 1) You can pay the registration fee at the official website using a credit card (VISA or Master card) when you register to the EARTH2013.
- 2) You can also pay the registration fee (handling charge) by the POS machine on-site in the conference registration day.
- 3) The accommodation fee and tour fee will be charged on-site by cash or credit card (VISA or Master card).

Cancellation and Refund:

- 1) Before 31st Aug., 2013: 70% of registration fee (minus bank transfer charge and other related expenses) would be refunded.
- 2) After 1st Sep., 2013: No refund.

If you apply for a refund, please E-mail to the Conference Secretariat to confirm the cancellation

of the registration and demand for a refund. At the same time, please inform us of your payment order number, name and remittance account details. We will refund your payment after the closure of EARTH2013 as soon as possible.

Visa

International participants: If you need a visa to China to attend the EARTH2013, please consult the Chinese Embassy in your country on the documents and fee to apply the visa as soon as possible. If a formal invitation letter is required, please contact the conference secretariat of EARTH2013. We will mail you the hard copy formal invitation letter in time and assist you to apply the visa.

Chinese Embassy in South Korea:

http://www.chinaemb.or.kr/kor/lshyw/lsk_qz/lq_sw/t234437.htm

Chinese Embassy in Japan:

<http://www.china-embassy.or.jp/jpn/lsfu/hzqzyw/t938310.htm>

Chinese Embassy in Thailand:

<http://www.chinaembassy.or.th/eng/lsfw/qzsk/t959329.htm>

Participants from Taiwan: You need to have the “mainland travel permit for Taiwan residents” (This can be also entrusted to travel agents in Taiwan.).

Participants from Hong Kong and Macao: You need to have the “Hong Kong and Macao residents traveling to the mainland passes”.

Invitation Letter

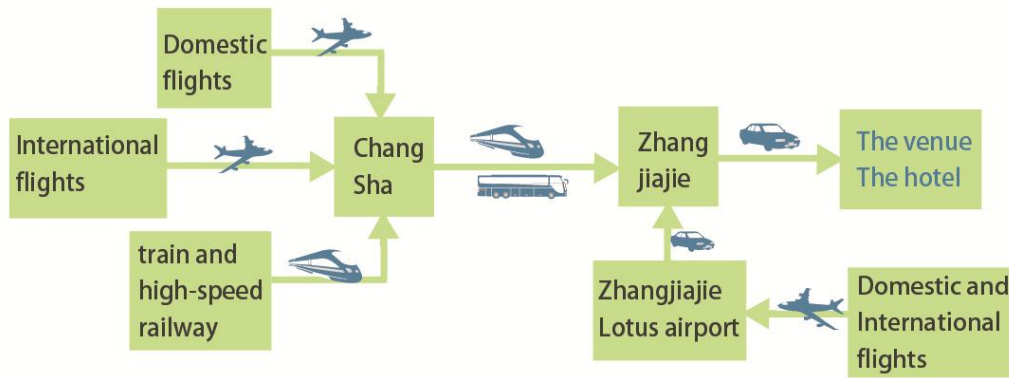
If a formal invitation letter is required, please contact the conference secretariat of EARTH2013. Please provide your name, address, zip code and phone number. The invitation letter will be sent in time.

Arrival

On the on-site registration day, we will shuttle participants at Zhangjiajie Lotus International Airport and Zhangjiajie Railway Station.

Travel to Zhangjiajie

The transportation to Zhangjiajie is very convenient. Zhangjiajie Lotus International Airport provides domestic flights to main cities, as well as direct international flights to Japan, Korea, Thailand and Taiwan. Through the Shichang Railway, Zhangjiajie is connected with the national railway network and has trains shuttling between Zhangjiajie and a number of large and medium-sized cities, such as Beijing, Shanghai, Guangzhou, Shenzhen, Changsha, Nanning, Guiyang, Yichang, Zhanjiang. You can also transit to Zhangjiajie from Changsha by coach. After arriving in Changsha by international flights, domestic flights, High Speed Rail or train, you can take train or coach to Zhangjiajie. We recommend coach by which it takes you 4 hours directly from Changsha Huanghua Airport, Changsha South Railway Station or Changsha Railway Station to Zhangjiajie. When you arrive in Zhangjiajie, you can take taxi to your hotel.



On-site registration

Time: 3rd Nov., 2013 from 13:00 to 18:00

Address: registration room of EARTH203 at Sunshine Hotel & Resort Zhangjiajie

Contracts: Prof. Zhihui Yang

Phone: +86-744-8558888

Fax: +86-744-8559999

Email: secretariat@earth2013.org

Map:



Accommodation

Please refer to the following hotels which are recommended by the EARTH2013. You may make a reservation in the EARTH2013 official website and pay it on-site when you arrive at the hotel.

1) Sunshine Hotel & Resort Zhangjiajie (Five-star)

Address: No. 2 Yongding East Ave., Yongding Dist., Zhanjiajie, Hunan province, China

Telephone: +86-744-8558888

Fax: + 86-744-8559999

E-mail: hotel@sunshinehotelzjj.com

Webpage: <http://www.sunshinehotelzjj.com/>

Quotation: (1) Superior Room: 498 RMB/night (two breakfasts)

Big single room or Double twin room

(2) Deluxe Room: 598 RMB/night (two breakfasts)

Big single room or Double twin room

2) Lan Tian Hotel (Four-star)

Address: No. 21 Chongwen Rd., Yongding Dist., Zhanjiajie, Hunan province, China

Telephone: +86-744-8278888

Fax: +86-744-8278199

Webpage: <http://www.lantianhotel.com/>

Quotation: Double twin Standard Room: 280 RMB/night (two breakfasts)

● **International Organizing Committee**

Chairman:

Liyuan Chai, Central South University

Members:

Junji Shibata, Kansai University

Hyungkyu Park, Korean Institute of Resources Recycling

Juu-En Chang, National Cheng Kung University

Dawan Wiwattanadate, Chulalongkorn University

Honorary Members:

Chuanfu Zhang, Central South University

Jae-Hyun Oh, Yonsei University

Min-Shing Tsai, National Cheng Kung University

● **Host Organizing Committee**

Chairman: Yi Kang, China National Nonferrous Metals Industry Association

Co-Chairmen: Kechao Zhou, Central South University

Members: Hongguo Zhang, Nonferrous Metal Society of China

Jie Li, Central South University

Houping Wu, Central South University

Huanwen Yang, Nonferrous Metal Society of China

Mingjun Teng, Central South University

Liyuan Chai, Central South University

Xueyan Yang, International Copper Association Ltd., China

Jiwei Wang, China Association of Resource Comprehensive Utilization

● **Technical Committee**

Chairman: Dingfan Qiu, Beijing General Research Institute of Mining and Metallurgy

Co-Chairmen: Yexiang Liu, Central South University

Yi Zhang, Institute of Process Engineering, Chinese Academy of Sciences

Tieyong Zuo, Beijing University of Technology

Chuanfu Zhang, Central South University

Members: Ansheng Feng, Zhengzhou Institute of Mineral Comprehensive Utilization,
Chinese Academy of Geological Sciences

Chenguang Bai, Chongqing University

Shenglian Luo, Hunan University

Shengming Xu, Tsinghua University

Jinhui Peng, Kunming University of Science and Technology

Tingsheng Qiu, Jiangxi University of Science and Technology

Zhancheng Guo, University of Science & Technology Beijing

Wangxi Zhu, National Natural Science Foundation of China

Liyuan Chai, Central South University

Xueyi Guo, Central South University

Bing Peng, Central South University

Yimin Zhang, Wuhan University of Science and Technology

● **Secretariat**

Secretary General: Zhihui Yang, Central South University

Huanwen Yang, Nonferrous Metal Society of China

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